

$$= (+80) + (-100) = -20 \text{ J} \quad (\text{B})$$

16 Ans: D

At point P, the displacement x is negative.

$$\text{Since } a = -\omega^2 x \Rightarrow F = ma = -m\omega^2 x.$$

A negative x will mean F is positive (B and C are eliminated as F is negative at these positions).

At point P, the velocity is negative.

Since $a - t$ graph is a cosine graph, $v - t$ graph will be a sine graph. Sketch this $v - t$ graph on the same t axis as the given $F - t$ graph. Can see that pt A has a positive velocity while pt D has a negative velocity. Hence ans is D.

